ME 333

MECHANICS OF MATERIALS

Lecture 11: Stress - Strain Relations II

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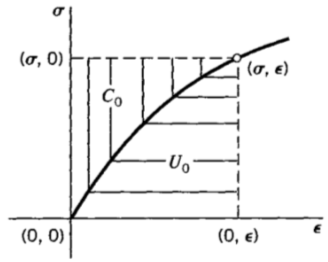
Strain Energy & Complementary Energy

- The strain energy and complementary energy can be defined as:

$$U_0 = \int \sigma \, d\epsilon \implies \sigma = \frac{dU_0}{d\epsilon}; \qquad C_0 = \int \epsilon \, d\sigma \implies \epsilon = \frac{dC_0}{d\sigma}$$

- Total area measure for a given stress strain pair is: $U_0 + C_0 = \sigma\epsilon$



Ref Fig: A. P. Boresi and R. J. Schmidt, Advanced Mechanics of Materials, Sixth Edition.

Strain Energy & Complementary Energy

- The net energy measure can be written as:

$$U_0 + C_0 = \sigma_{xx}\epsilon_{xx} + \sigma_{yy}\epsilon_{yy} + \sigma_{zz}\epsilon_{zz} + 2\sigma_{xy}\epsilon_{xy} + 2\sigma_{yz}\epsilon_{yz} + 2\sigma_{zx}\epsilon_{zx}$$

- In case of a *linear elastic material*, the strain energy density equals complementary energy density, i.e.:

$$U_0 = C_0$$

- Taking the partial derivative for stress components with respect to ϵ_{xx} on both sides:

$$2\frac{\partial U_0}{\partial \epsilon_{xx}} = \frac{\partial \sigma_{xx}}{\partial \epsilon_{xx}}\epsilon_{xx} + \sigma_{xx} + \frac{\partial \sigma_{yy}}{\partial \epsilon_{xx}}\epsilon_{yy} + \frac{\partial \sigma_{zz}}{\partial \epsilon_{xx}}\epsilon_{zz} + 2\frac{\partial \sigma_{xy}}{\partial \epsilon_{xx}}\epsilon_{xy} + 2\frac{\partial \sigma_{yz}}{\partial \epsilon_{xx}}\epsilon_{yz} + 2\frac{\partial \sigma_{zx}}{\partial \epsilon_{xx}}\epsilon_{zx}$$

- Recall that:

$$\sigma_{xx} = \frac{\partial U_0}{\partial \epsilon_{xx}}, \; .... \; , \sigma_{xy} = \frac{1}{2}\frac{\partial U_0}{\partial \epsilon_{xy}}, \; ....$$

Stress - Strain Mapping

- Substituting into the equation for the total energy density, we have:

$$\sigma_{xx} = \frac{\partial^2 U_0}{\partial \epsilon_{xx}\partial \epsilon_{xx}}\epsilon_{xx} + \frac{\partial^2 U_0}{\partial \epsilon_{yy}\partial \epsilon_{xx}}\epsilon_{yy} + \frac{\partial^2 U_0}{\partial \epsilon_{zz}\partial \epsilon_{xx}}\epsilon_{zz} + \frac{\partial^2 U_0}{\partial \epsilon_{xy}\partial \epsilon_{xx}}\epsilon_{xy} + \frac{\partial^2 U_0}{\partial \epsilon_{yz}\partial \epsilon_{xx}}\epsilon_{yz} + \frac{\partial^2 U_0}{\partial \epsilon_{zx}\partial \epsilon_{xx}}\epsilon_{zx}$$

Different Line of Thought -

- Stress can be mapped to strain as:

$$\boldsymbol{\sigma} = \mathbb{C}\,\boldsymbol{\epsilon}$$

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} = \mathbb{C} \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix}$$

Stress - Strain Mapping

- Voigt Notation:

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \\ \sigma_{yx} \\ \sigma_{zx} \\ \sigma_{zy} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} & C_{17} & C_{18} & C_{19} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} & C_{27} & C_{28} & C_{29} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} & C_{37} & C_{38} & C_{39} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} & C_{47} & C_{48} & C_{49} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} & C_{57} & C_{58} & C_{59} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} & C_{67} & C_{68} & C_{69} \\ C_{71} & C_{72} & C_{73} & C_{74} & C_{75} & C_{76} & C_{77} & C_{78} & C_{79} \\ C_{81} & C_{82} & C_{83} & C_{84} & C_{85} & C_{86} & C_{87} & C_{88} & C_{89} \\ C_{91} & C_{92} & C_{93} & C_{94} & C_{95} & C_{96} & C_{97} & C_{98} & C_{99} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{xz} \\ \epsilon_{yz} \\ \epsilon_{yx} \\ \epsilon_{zx} \\ \epsilon_{zy} \end{Bmatrix}$$

- Number of constants is 81

Stress - Strain Mapping

- Voigt Notation:

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{xz} \\ \epsilon_{yz} \end{Bmatrix}$$

- Number of constants is 36
- Expanding and writing the first term we get:

$$\sigma_{xx} = C_{11}\epsilon_{xx} + C_{12}\epsilon_{yy} + C_{13}\epsilon_{zz} + C_{14}\epsilon_{xy} + C_{15}\epsilon_{xz} + C_{16}\epsilon_{yz}$$

Stress - Strain Mapping

- Voigt Notation: $\boldsymbol{\sigma} = \mathbb{C}\,\boldsymbol{\epsilon}$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{xz} \\ \epsilon_{yz} \end{Bmatrix}$$

- Comparing

$$C_{11} = \frac{\partial^2 U_0}{\partial \epsilon_{xx} \partial \epsilon_{xx}}; \quad C_{12} = \frac{\partial^2 U_0}{\partial \epsilon_{xx} \partial \epsilon_{yy}} \quad \cdots$$

where:

$$\sigma_{xx} = \frac{\partial U_0}{\partial \epsilon_{xx}}, \; .... \; , \sigma_{xy} = \frac{1}{2}\frac{\partial U_0}{\partial \epsilon_{xy}}, \; ....$$

Stress - Strain Mapping

- Elements of the stiffness matrix are:

$$C_{11} = \frac{\partial^2 U_0}{\partial \epsilon_{xx} \partial \epsilon_{xx}}; \quad C_{12} = \frac{\partial^2 U_0}{\partial \epsilon_{xx} \partial \epsilon_{yy}} \quad \cdots$$

- Two kinds of symmetries exist above.

Stress - Strain Mapping

- Due to symmetry of stiffness tensor:

$$\begin{Bmatrix}\sigma_{xx}\\\sigma_{yy}\\\sigma_{zz}\\\sigma_{xy}\\\sigma_{xz}\\\sigma_{yz}\end{Bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16}\\ C_{12} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26}\\ C_{13} & C_{23} & C_{33} & C_{34} & C_{35} & C_{36}\\ C_{14} & C_{24} & C_{34} & C_{44} & C_{45} & C_{46}\\ C_{15} & C_{25} & C_{35} & C_{45} & C_{55} & C_{56}\\ C_{16} & C_{26} & C_{36} & C_{46} & C_{56} & C_{66}\end{bmatrix}\begin{Bmatrix}\epsilon_{xx}\\\epsilon_{yy}\\\epsilon_{zz}\\\epsilon_{xy}\\\epsilon_{xz}\\\epsilon_{yz}\end{Bmatrix}$$

- Number of constants is 21

Stress - Strain Mapping

- In most materials, normal stresses are dependent on normal strains and shear stresses on shear strains only.

$$\begin{Bmatrix}\sigma_{xx}\\\sigma_{yy}\\\sigma_{zz}\\\sigma_{xy}\\\sigma_{xz}\\\sigma_{yz}\end{Bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{13} & 0 & 0 & 0\\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0\\ C_{13} & C_{23} & C_{33} & 0 & 0 & 0\\ 0 & 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & C_{55} & 0\\ 0 & 0 & 0 & 0 & 0 & C_{66}\end{bmatrix}\begin{Bmatrix}\epsilon_{xx}\\\epsilon_{yy}\\\epsilon_{zz}\\\epsilon_{xy}\\\epsilon_{xz}\\\epsilon_{yz}\end{Bmatrix}$$

- Number of constants is 9.
- Materials which require 9 constants to fully characterize the material response are called “*Orthotropic Materials*”

Stress - Strain Mapping

- If the material has two orthogonal planes about which the properties (say x and y) are same while the properties along the third orthogonal direction (z) differ.

$$\begin{Bmatrix}\sigma_{xx}\\\sigma_{yy}\\\sigma_{zz}\\\sigma_{xy}\\\sigma_{xz}\\\sigma_{yz}\end{Bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{13} & 0 & 0 & 0\\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0\\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0\\ 0 & 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & C_{55} & 0\\ 0 & 0 & 0 & 0 & 0 & C_{55}\end{bmatrix}\begin{Bmatrix}\epsilon_{xx}\\\epsilon_{yy}\\\epsilon_{zz}\\\epsilon_{xy}\\\epsilon_{xz}\\\epsilon_{yz}\end{Bmatrix}$$

- Number of constants is 5.
- Materials which require 5 constants to fully characterize the material response are called “*Transversely Isotropic Materials*”

Stress - Strain Mapping

- If the material response is independent of the direction of loading then:

$$\begin{Bmatrix}\sigma_{xx}\\\sigma_{yy}\\\sigma_{zz}\\\sigma_{xy}\\\sigma_{xz}\\\sigma_{yz}\end{Bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0\\ 0 & 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & C_{44} & 0\\ 0 & 0 & 0 & 0 & 0 & C_{44}\end{bmatrix}\begin{Bmatrix}\epsilon_{xx}\\\epsilon_{yy}\\\epsilon_{zz}\\\epsilon_{xy}\\\epsilon_{xz}\\\epsilon_{yz}\end{Bmatrix}$$

- Number of constants is 2. Here C_{44} is a function of C_{11} and C_{12}.
- Materials which require 2 constants to fully characterize the material response are called “*Isotropic Materials*”

Stress - Strain Mapping

$\boxed{Q}$ How are the C's related to material properties E and ν ?

Isotropic Materials

- In case of isotropic materials, considering only principal stresses and strains, the strain energy density can be written as:

$$U_0 = \frac{1}{2}C_{11}\epsilon_1^2 + \frac{1}{2}C_{12}\epsilon_1\epsilon_2 + \frac{1}{2}C_{13}\epsilon_1\epsilon_3 + \frac{1}{2}C_{12}\epsilon_1\epsilon_2 + \frac{1}{2}C_{22}\epsilon_2^2 + \frac{1}{2}C_{23}\epsilon_2\epsilon_3 + \frac{1}{2}C_{13}\epsilon_1\epsilon_3 + \frac{1}{2}C_{23}\epsilon_2\epsilon_3 + \frac{1}{2}C_{33}\epsilon_3^2$$

- In case of isotropic materials:

$$C_{11} = C_{22} = C_{33} = C_1$$
$$C_{12} = C_{23} = C_{31} = C_2$$

- The strain energy density can then be recast in terms of C_1 and C_2 as:

$$U_0 = \frac{1}{2}C_2\left(\epsilon_1 + \epsilon_2 + \epsilon_3\right)^2 + \frac{C_1 - C_2}{2}\left(\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2\right)$$

Isotropic Materials

- Let us introduce two material parameters:

$$\lambda = C_2; \qquad \mu = G = \frac{C_1 - C_2}{2}$$

- The strain energy density can then be written as:

$$\begin{aligned} U_0 &= \frac{\lambda}{2}\left(\epsilon_1 + \epsilon_2 + \epsilon_3\right)^2 + \mu\left(\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2\right) \\ &= \left(\frac{\lambda}{2} + \mu\right)\bar{I}_1^2 - 2\mu\bar{I}_2 \end{aligned}$$

- The strain energy density in the most generalized scenario of loading can be written as:

$$U_0 = \frac{1}{2}\lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz})^2 + \mu(\epsilon_{xx}^2 + \epsilon_{yy}^2 + \epsilon_{zz}^2 + 2\epsilon_{xy}^2 + 2\epsilon_{yz}^2 + 2\epsilon_{xz}^2)$$

Hooke's Law

- Differentiating with respect to components of strain, we obtain:

$$\sigma_{xx} = \frac{\partial u}{\partial \epsilon_{xx}} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{xx}$$
$$\sigma_{yy} = \frac{\partial u}{\partial \epsilon_{yy}} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{yy}$$
$$\sigma_{zz} = \frac{\partial u}{\partial \epsilon_{zz}} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{zz}$$
$$\sigma_{xy} = 2G\epsilon_{xy}; \quad \sigma_{yz} = 2G\epsilon_{yz}; \quad \sigma_{zx} = 2G\epsilon_{zx}$$

- Inverting we obtain the Hooke's Law as:

$$\epsilon_{xx} = \frac{\lambda + G}{G(3\lambda + 2G)}\left[\sigma_{xx} - \frac{\lambda}{2\lambda + G}(\sigma_{yy} + \sigma_{zz})\right]$$

Stress - Strain Mapping

$\boxed{Q}$ What is Poisson Ratio ?

Hooke's Law

[Example] Uniaxial tension test (along z)

- $\sigma_{zz} = \sigma$ = Constant, rest are zero.
- Using the relation for strains from above:

$$\epsilon_{xx} = \epsilon_{yy} = -\frac{\lambda\sigma}{2G(3\lambda + 2G)}; \qquad \epsilon_{zz} = \frac{\sigma(\lambda + G)}{G(3\lambda + 2G)}$$

- Poisson's ratio is defined as:

$$\nu = -\frac{\epsilon_{xx}}{\epsilon_{zz}} = \frac{\lambda}{2(\lambda + G)}$$

- Based on definition of normal strain along z:

$$\epsilon_{zz} = \frac{\sigma}{E} \implies E = \frac{G(3\lambda + 2G)}{(\lambda + G)}$$

Hooke's Law

- Writing in matrix form: $\quad \boldsymbol{\epsilon} = \mathbb{S}\,\boldsymbol{\sigma}$

$$\begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{xz} \\ \epsilon_{yz} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & -\frac{\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2G} \end{bmatrix} \begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{xz} \\ \sigma_{yz} \end{Bmatrix}$$

- Inverting we obtain strains in terms of stresses as:

$$\epsilon_{xx} = \frac{1}{E}\left[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})\right]$$

Hooke's Law

- Solving for λ and G in terms of E and ν, we obtain:

$$\lambda = \frac{\nu E}{[(1+\nu)(1-2\nu)]}; \qquad G = \frac{E}{2(1+\nu)}$$

- Stresses can then be written in terms of strains as:

$$\sigma_{xx} = \frac{E(1-\nu)}{[(1+\nu)(1-2\nu)]}\epsilon_{xx} + \frac{\nu E}{[(1+\nu)(1-2\nu)]}(\epsilon_{yy} + \epsilon_{zz})$$

- One may use the stress strain relations to also obtain relationship between the stress and strain invariants in terms of either Lame's constants or Young's Modulus and Poisson ratio.

Reading Assignment

Chapter 3: Section 3.1 - 3.3.2